

TECHNICAL PUBLICATION PH009 – YASKAWA V1000 SETUP PROCEDURE

Heat and Control Product Handling Systems

Yaskawa V1000 Inverter Parameter Values for Synchronized SwitchBack™ 260E Conveyors			
V1000 Parameter Number	Lead Conveyor	Follower Conveyor	Function Description
L3-01	0	0	Stall Prevention accel selection (0=Disabled)
L3-04	0	0	Stall prevention during decel is disabled.
L3-05	0	0	Stall Prevention running selection (0=Disabled)
L8-35	0.142 (changes when E2-11 changes)	0.142 (changes when E2-11 changes)	Installation method.
O3-02	1	1	Enables reading of parameter settings. For handheld LCD operator only.
DIP Switch Setup			
S1	V – [I] Current selected (right side)	[V] – I Voltage selected (left side)	Voltage (0 to 10 V) or current (4 to 20 mA or 0 to 20 mA: Default) input.
S2	[OFF] – ON off selected (left side)	[OFF] – ON off selected (left side)	Set main frequency reference.
S3	[SINK] Sink selected (upper) SOURCE	[SINK] Sink selected (upper) SOURCE	Sinking mode: Default = 0 V common Sourcing mode = +24 V common
Mechanical Setup			
With the key slot of the taper lock bushing of the main drive sprocket in the 12 o'clock position, the hole in the sprocket should be roughly in the 9 o'clock position.			

Table PH009-9 – Yaskawa V1000 Inverter Parameter Values for Synchronized SwitchBack™ 260E Conveyors

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Revision Record

Revision Level	Date	By	Description
Revision 08	10/12/15	Ken Petri, Tom Knodell & Arlene Mason	• Added new tables and revised certain parameters. • Added Revision Record • Added Table of Contents • Added Troubleshooting section.
Revision 09	3/14/16	Ken Petri, Tom Knodell & Arlene Mason	• Removed obsolete 200A parameters and added 260E-G3 / 260E-G3 Plus parameters. • Revised 90E to include 1/3 HP motor amps.
Revision 10	1/26/17	Tom Knodell & Arlene Mason	• Added parameters for blending systems. • Removed obsolete V7 parameters for Dust Collection

Technical Publication PH015

Allen-Bradley

PowerFlex 525 Setup

Parameters

Rev. 07



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Important User Information



Attention: In ALL high voltage (380V-480V) FastBack and Revolution applications, use a line/load reactor placed by the motor when the motor leads are known or suspected to be greater than 50ft. See Table PH015-1 for the specific line/load reactor by application.

For motor leads greater than 200ft, the recommended solutions below are inadequate. Consult Heat and Control, Inc., Product Handling Systems at (469) 464-2700 before installing the equipment.

FastBack Application	Heat and Control Part No.	Description
400V Class		
FB 260E-G1 & -G2	20537331	Load Reactor, 3PH, 12mH
FB 50E/90E	20537332	Load Reactor, 3PH, 32mH
Rev-Gate		
Rev-Tumble Drum		
600V Class		
FB 260E-G1 & -G2	20544966	Load Reactor, 3PH, 20mH

Table PH015-1 – Load Reactors

Setup Procedure for Allen-Bradley PowerFlex 525

This procedure describes how to set or restore the Heat and Control factory parameter settings for the **Allen-Bradley PowerFlex 523/525** and how to read and copy those parameters to other drives of the same power and voltage ratings. If following these steps does not correct the problem, call the Heat and Control, Inc. Service Office in Lititz, PA at (717) 519-5240 or (800) 648-5313.

Initial Setup

For initial setup of the PowerFlex 525, you will require the following tools:

- Allen-Bradley PowerFlex 525 Human Interface Module (22-HIM-A3)
(Part no. 20546184)

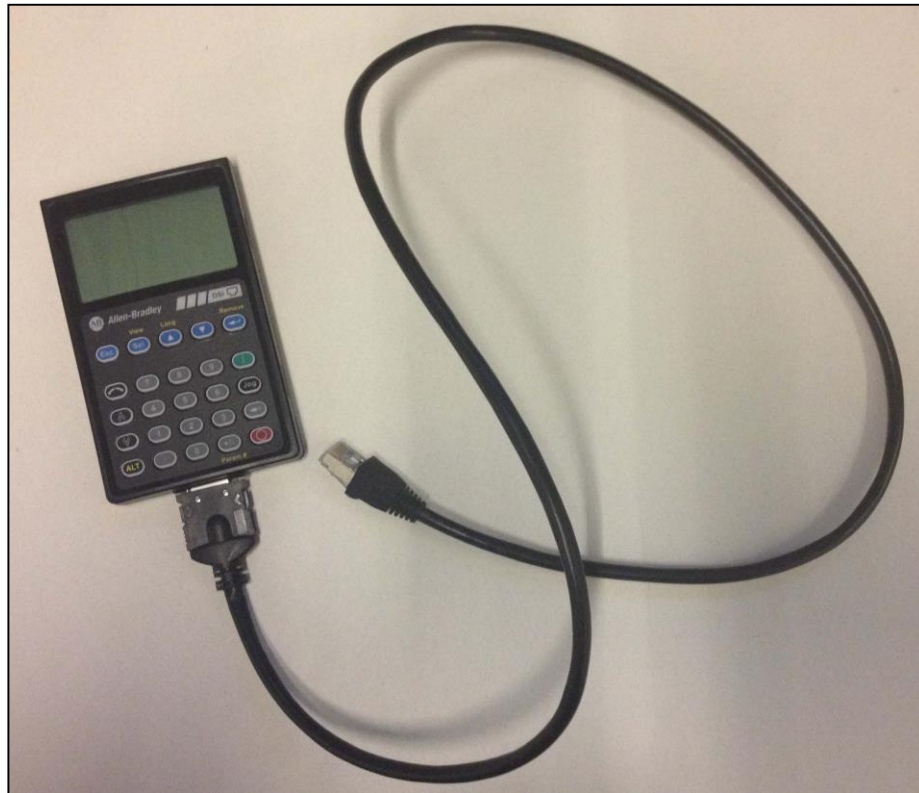


Photo PH015-1 – Required Tools

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Using the Hand-Held LCD Human Interface Machine (HIM)

The LCD HIM will allow you to:

- Read or modify drive parameters.
- Copy drive parameter settings to another Allen-Bradley drive.
- Operate the drive.
- Monitors drive operation status.

The LCD HIM can be connected to the drive one of two ways:

- By plugging the RJ45 connector of the HIM Interface cable into DSI Port at the center of the drive (see Photo PH015-2).
- By plugging the RJ45 connector into the external communication port on the FastBack® (see Photo PH015-3 and Photo PH015-4).

The latter is the preferred method.



Photo PH015-2 – HIM Connection

1. Supply Power to the FastBack® (or FDX)

Connect power to the FastBack® by rotating the emergency disconnect switch to the ON position.

Enable power provided to the FastBack® system at the source by turning on the power at the power distribution panel.



Ensure the correct power voltage is supplied to the FastBack®.

To prevent the FastBack® from accidentally starting during this procedure, lockout / tag out before servicing.

Moving parts are located in a separate compartment behind the accessory panel. Do not operate with chassis cover removed. Exercise caution if turning the mechanisms by hand to avoid crushing or pinching injuries.

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2. Connecting the HIM

Locate the communication port behind the FastBack® Accessory Panel (see Photo PH015-3).

Remove the port cap by turning it counterclockwise.

Plug the operator communication cable into the port by depressing the locking tab on the connector, pushing straight in and releasing the tab (see Photo PH015-4).

Once connected, the display will initialize then show the home screen (see Photo PH015-5).



Photo PH015-3 – External Communication Ports

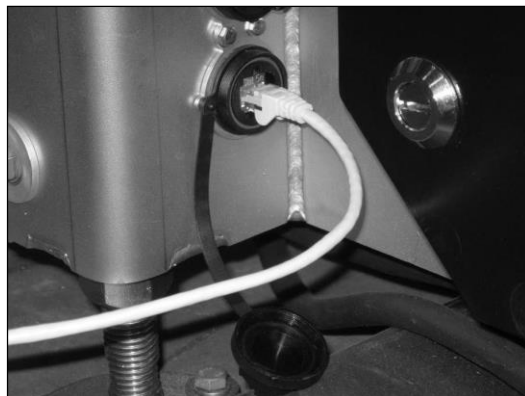


Photo PH015-4 – Communication Cable Plugged Into Port



Photo PH015-5 – PowerFlex 525 HIM Display

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3. Set and View Parameter Values

Step	Key(s)	Example Display
1. When power is applied, the last user-selected Basic Display Group parameter number is briefly displayed with flashing characters. The display then defaults to that parameter's current value (Example shows the value of b001 [Output Freq] with the drive stopped).		
2. Press Esc to display the Basic Display Group parameter number shown on power-up. The parameter number will flash.		
3. Press Esc to enter the parameter group list. The parameter group letter will flash.		
4. Press the Up Arrow or Down Arrow to scroll through the group list (b, P, t, C, L, d, A, f, N, M, and Gx).	or	
5. Press Enter or Sel to enter a group. The right digit of the last viewed parameter in that group will flash.	or	
6. Press the Up Arrow or Down Arrow to scroll through the parameter list.	or	
7. Press Enter to view the value of the parameter. Or Press Esc to return to the parameter list.		
8. Press Enter or Sel to enter Program Mode and edit the value. The right digit will flash and the word Program on the LCD display will light up.	or	
9. Press the Up Arrow or Down Arrow to change the parameter value.	or	
10. If desired, press Sel to move from digit to digit or bit to bit. The digit or bit that you can change will flash.		
11. Press Esc to cancel a change and exit Program Mode. Or Press Enter to save a change and exit Program Mode. The digit will stop flashing and the word Program on the LCD display will turn off.	or	
12. Press Esc to return to the parameter list. Continue to press Esc to back out of the programming menu. If pressing Esc does not change the display, then b001 [Output Freq] is displayed. Press Enter or Sel to enter the group list again.		

4. Copy Parameter Values

If more than one drive is to be initialized, copy the values from the programmed drive to the HIM using the steps below:

- From the Main Menu, press **Sel** to get to folders, then press **←** until at the “MEM” folder.
- Use **▲** or **▼** to highlight “HIM CopyCat”. Press.
- Use **▲** or **▼** to highlight “Device -> HIM”.
- Press **↵** to copy values to the HIM.
- A dialogue box will appear allowing you to name the saved set of parameters (Figure PH010-6). Use **▲** or **▼** to highlight and **Sel** to select letters for the name.
- When complete, press **↵**.



Photo PH015-6 – Device -> HIM Dialogue Box

- A bar will appear showing the percentage of save completed. Once 100% complete, another dialogue box will appear stating “Saving of File (*filename*) to HIM is Complete!”.
- Press **Esc** three times to return to the Main Menu.

With the required parameter values saved to the HIM, copy the values from the HIM to another PowerFlex 525 drive using the steps below:

- From the Main Menu, press **Sel** to get to folders, then press **←** until at the “MEM” folder.
- Use **▲** or **▼** to highlight “HIM CopyCat”. Press **↵**.
- Use **▲** or **▼** to highlight “Device <- HIM”.

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- d. Press  after selecting file to load, to copy values to the drive.
- e. A dialogue box will appear requesting confirmation. Press .
- f. Once 100% complete, another dialogue box will appear stating "Saving of File (*filename*) to Device is Complete!".
- g. Press  three times to return to the Main Menu.

5. Restore Factory Default Values

If required, you can restore all factory default values for drive parameters one of two ways:

Method 1

- a. From the Main Menu, press  to get to folders, then press  until at the "MEM" folder.
- b. Use  or  to highlight "Reset To Default".
- c. Press  to reset values in the drive.
- d. When complete, press  three times to return to the Main Menu.

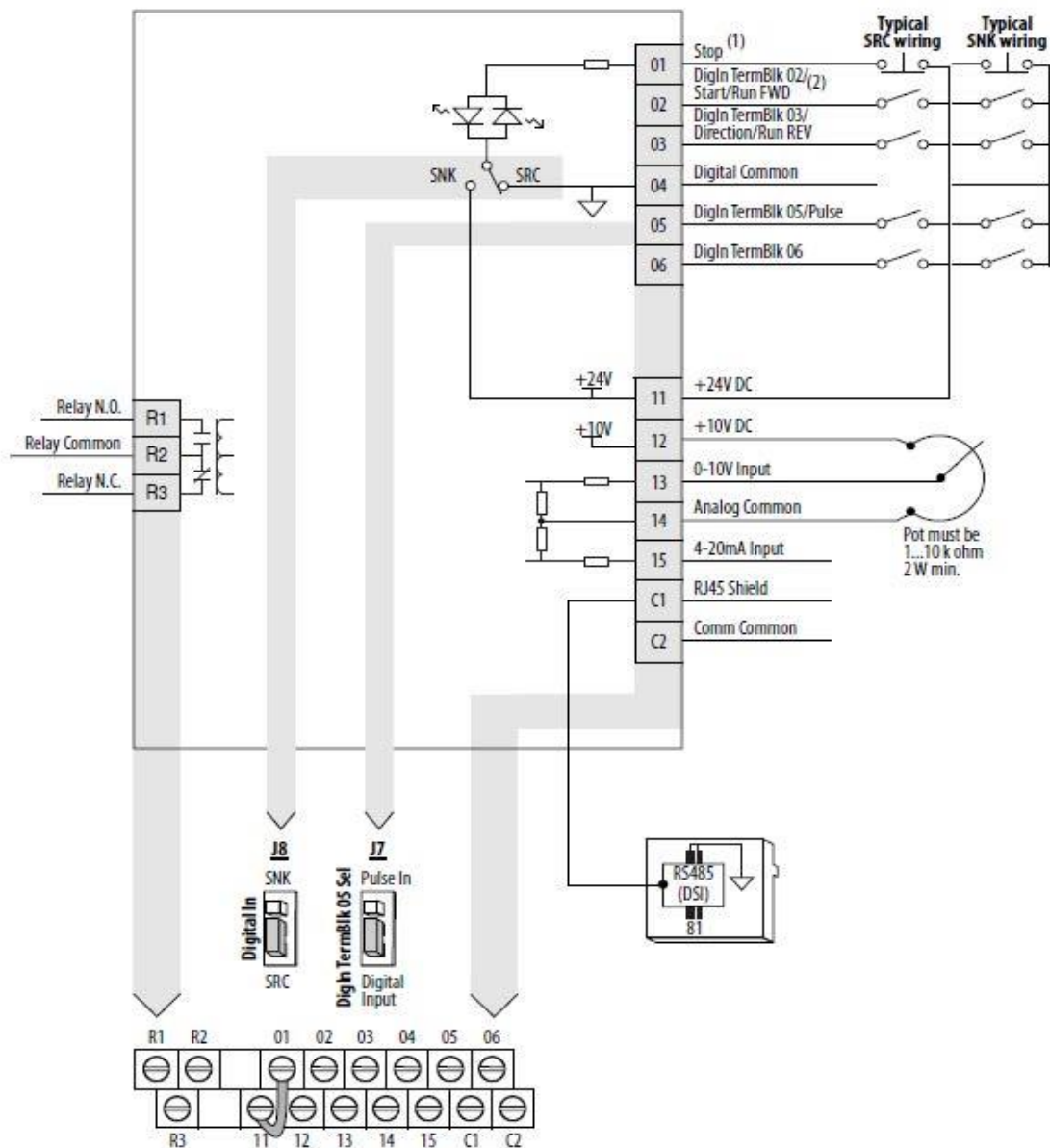
Method 2

- a. From the Main Menu, press , release it, then press  to select the Parameter screen.
- b. Use the numeric keypad to select Parameter P053. Press  twice.
- c. Use ,  or the numeric keypad to change the value to "2 – Factory Reset".
- d. Press  three times to return to the Main Menu.

Drive I/O Wiring Diagram

Control Terminal Block

PowerFlex 523 Control I/O Wiring Block Diagram



(1)

IMPORTANT I/O Terminal 01 is always a stop input. The stopping mode is determined by the drive setting.

The drive is shipped with a jumper installed between I/O Terminals 01 and 11. Remove this jumper when using I/O Terminal 01 as a stop or enable input.

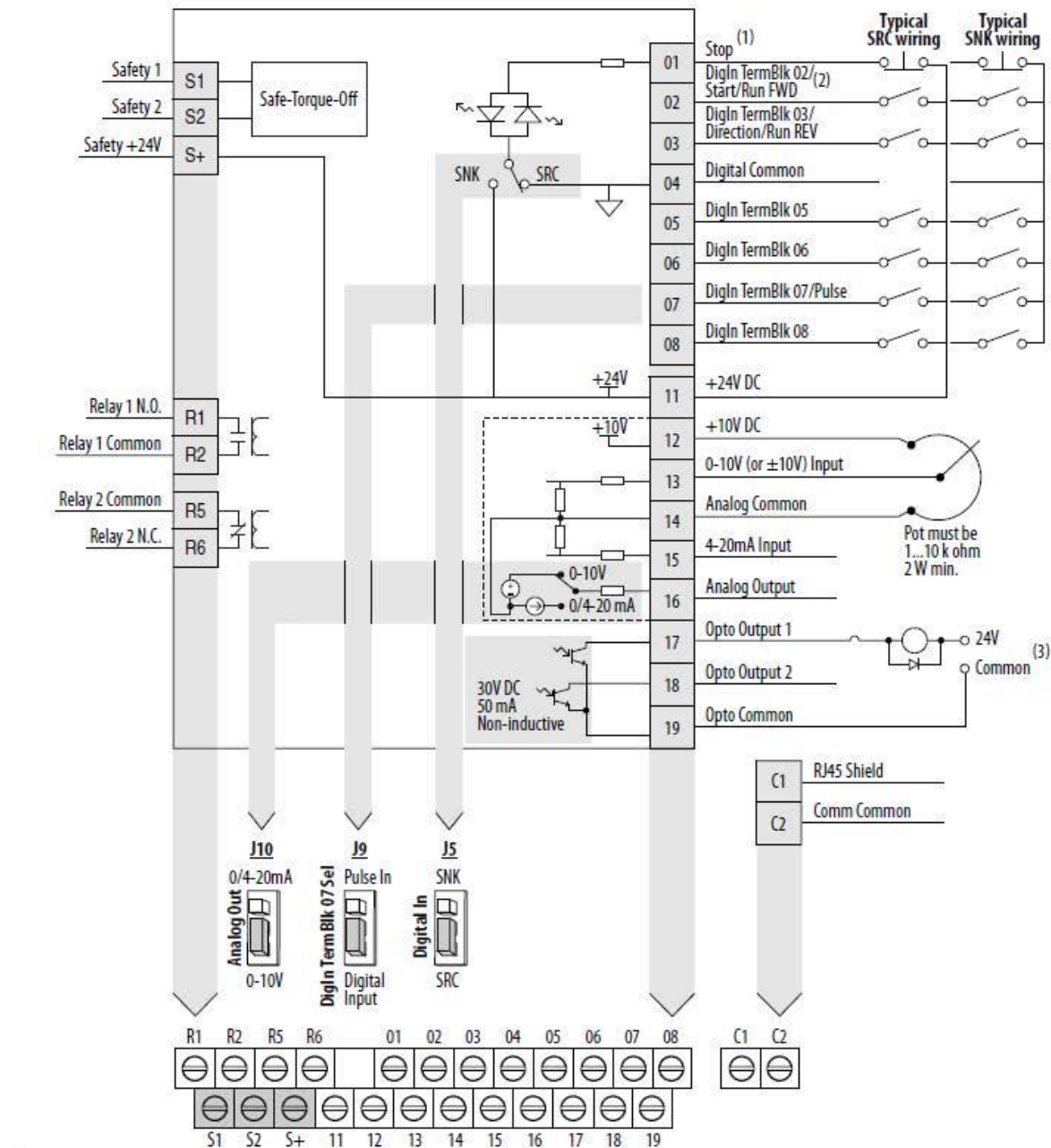
- (2) Two wire control shown. For three wire control use a momentary input  on I/O Terminal 02 to command a start. Use a maintained input  for I/O Terminal 03 to change direction.
- (3) Only one analog frequency source may be connected at a time. If more than one reference is connected at the same time, an undetermined frequency reference will result.

Figure PH015-1 – PowerFlex 523 Control I/O Wiring Block Diagram

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PowerFlex 525 Control I/O Wiring Block Diagram



(1)

IMPORTANT I/O Terminal 01 is always a stop input. The stopping mode is determined by the drive setting. The drive is shipped with a jumper installed between I/O Terminals 01 and 11. Remove this jumper when using I/O Terminal 01 as a stop or enable input.

(2) Two wire control shown. For three wire control use a momentary input on I/O Terminal 02 to command a start. Use a maintained input for I/O Terminal 03 to change direction.

(3) Only one analog frequency source may be connected at a time. If more than one reference is connected at the same time, an undetermined frequency reference will result.

Figure PH015-2 – PowerFlex 525 Control I/O Wiring Block Diagram

Drive Specific Information

FastBack® FDX Conveyors

The following is a typical control I/O schematic diagram and drive settings for a FastBack® FDX Conveyor.

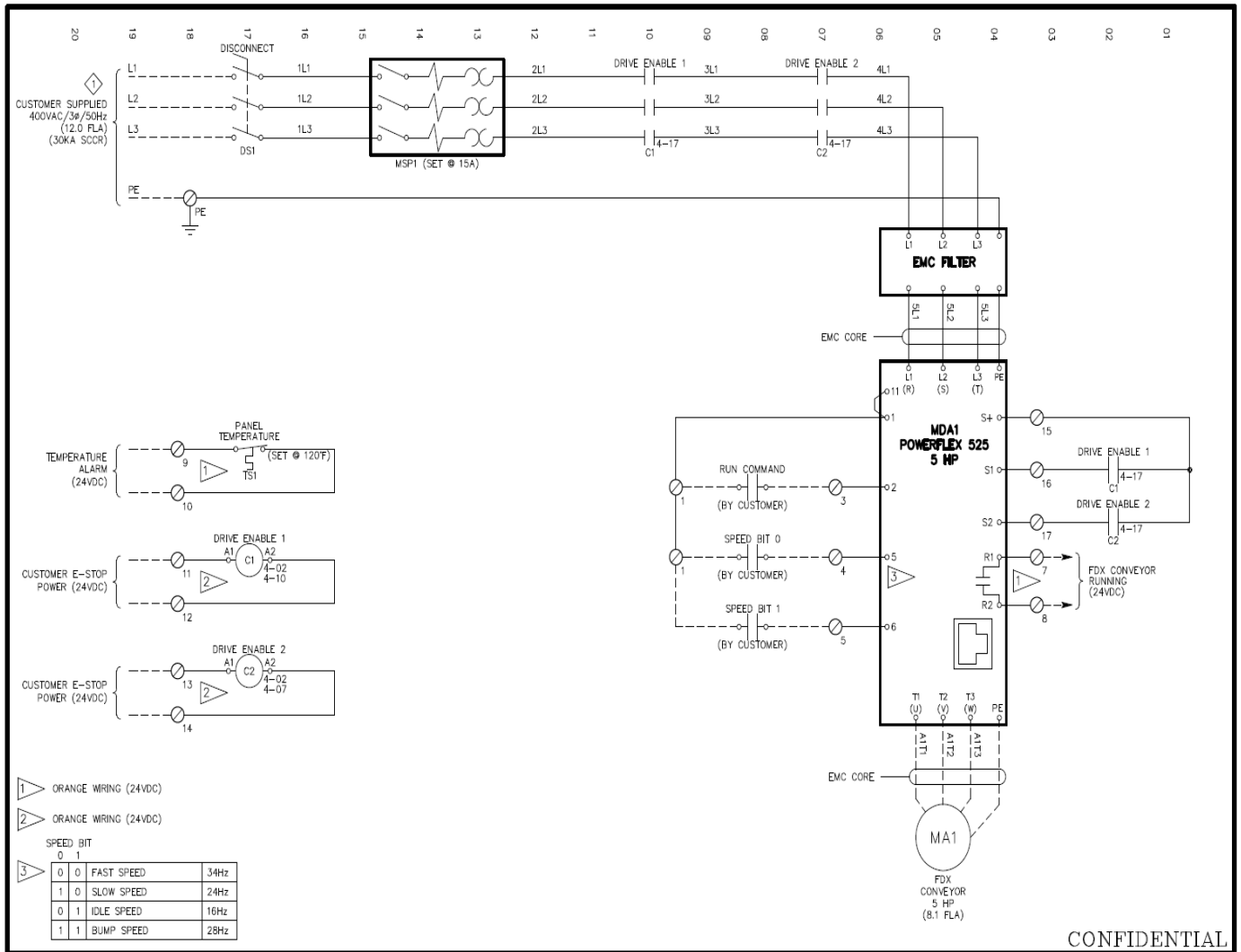


Figure PH015-3 – PowerFlex 525 Typical FDX 2-Wired Start Schematic Diagram

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Allen-Bradley PowerFlex 525 VFD Parameter Values for FastBack® FDX Conveyors

PF-523/525 Parameter Number	Parameter Name	FDX Values	Functional Description
P030	Language	1 "English"	Selects the display language when using an LCD HIM. This parameter is not functional with an LED HIM. Default = 1 "English"
P031	Motor NP Volts	208 OR 230 OR 400 OR 460	Set to line voltage.
P032	Motor NP Hertz	60	Set to motor nameplate rated frequency.
P033	Motor OL Current	13.0 (230 VAC) OR 6.5 (460 VAC)	Sets the motor nameplate overload current
P034	Motor NP FLA	13.0 (230 VAC) OR 6.5 (460 VAC)	Use 230 values for 190-240 VAC applications. Use 460 values for 380-500 VAC applications Set to motor nameplate rated full load amps.
P035	Motor Poles	4	Defines the number of poles in the motor. Default = 4
P036	Motor NP RPM	1800	Set to motor nominal RPM.
P037	Motor NP Power	3.75 (5HP)	Set to motor nameplate rated power. (kW)
P039	Torque Perf Mode	1	The method of motor control used in the drive. 0 = V/Hz 1 = "SVC" (Default) 2 = "Economize" 3 = "Vector"
P041	Accel Time 1	3.0	Sets the rate of accel 1 for all speed increases. Default = 10.0s.
P042	Decel Time 1	3.0	Sets the rate of decel 1 for all speed decreases. Default = 10.0s.
P043	Minimum Freq	10	Sets the lowest frequency the drive will output.
P044	Maximum Freq	42	Sets the highest frequency the drive will output.
P045	Stop Mode	0	Determines the stopping mode used by the drive when a stop is initiated. 0 = Ramp, CF (Default) 1 = Coast, CF 2 = DC Brake, CF
P046	Start Source 1	2 (Hardwired) OR 5 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Keypad (Default) 2 = DigIn TrmBlk 3 = Serial/DSI 4 = Network Opt. 5 = Ethernet/IP
P047	Speed Ref 1	7 (Hardwired) OR 15 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Drive Pot (Default) 6 = 4-20 mA 7 = Preset Frequency 15 = Ethernet/IP
A410	Preset Freq 0	30	Provides an internal fixed speed command value. Default = 0 Hz. Used as "Fast" Speed Note: uses Accel 1 & Decel 1
A411	Preset Freq 1	20	Provides an internal fixed speed command value. Default = 5 Hz. Used as "Slow" speed Note: uses Accel 1 & Decel 1
A412	Preset Freq 2	10	Provides an internal fixed speed command value. Default = 10Hz. Used as "Idle" speed Note: uses Accel 2 & Decel 2

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PF-523/525 Parameter Number	Parameter Name	FDX Values	Functional Description
A413	Preset Freq 3	28	Provides an internal fixed speed command value. Default = 20Hz. Note: uses Accel 2 & Decel 2
A440	PWM Frequency	2 kHz	Sets the carrier frequency for the PWM output. Drive derating may occur at higher carrier frequencies. Default = 4kHz.
A442	Accel Time 2	3.0	Sets the rate of accel 2 for all speed increases. Default = 10.0s.
A443	Decel Time 2	3.0	Sets the rate of decel 2 for all speed decreases. Default = 10.0s.
A484	Current Lmt Val	19.5 (230 VAC) OR 9.7 (460 VAC)	Maximum output current allowed before current limiting occurs. Choose the value for the voltage in the application
A543	Start At PowerUp	1	Enables/disables a feature to issue a Start or Run command and automatically resume running at commanded speed after drive input power is restored. Default = 0 "Disabled"
A544	Reversed Disable	1	Enables/disables the function that allows the direction of motor rotation to be changed. 0 = Rev. Enabled (Default) 1 = Rev. Disabled
T062	DigIn TermBlk 02	48	Programmable Digital Input. 0 = Not Used 48 = 2-Wire Forward (Default) 49 = 3-Wire Start
T065	DigIn TermBlk 05	7	Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T066	DigIn TermBlk 06	7	Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T076	Relay Out1 Sel	2	Determines the operation of the programmable output relay. 0 = Ready/Fault (Default) 1 = At Frequency 2 = Motor Running 3 = Reverse 4 = Motor Overload
T105	Safety open En	1 (normal) OR 0 (if required by customer)	Sets the action when both safety inputs (Safety 1 and Safety 2) are disabled (de-energized – no power is applied). 0 = Fault Enable (Default) 1 = Fault Disable

IF **ETHERNET** IS USED SEE APPENDIX A FOR PARAMETERS.

Table PH015-2 – PowerFlex 525 VFD Parameters for FastBack® FDX Conveyor

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The following is a typical control I/O schematic diagram and drive settings for a FastBack® Conveyor.



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Allen-Bradley PowerFlex 525 VFD Parameter Values for FastBack® Conveyors

PF-523/525 Parameter Number	Parameter Name	50E / 90E Values		260E / 260E-G2 / 260E-G3 Values (1 HP Motor)	260E-G3 Plus Values (2 HP Motor)	Functional Description
P030	Language	1 "English"				Selects the display language when using an LCD HIM. This parameter is not functional with an LED HIM. Default = 1 "English"
P031	Motor NP Volts	208 OR 230 OR 400 OR 460				Set to line voltage
P032	Motor NP Hertz	60				Set to motor nameplate rated frequency.
P033	Motor OL Current	<u>1/3 HP</u> 2.6 (230V) OR 1.3 (460V)	<u>1/4 HP</u> 2.2 (230V) OR 1.2 (460V)	4.2 (230 VAC) OR 2.1 (460 VAC) OR 1.9 (575 VAC)	8.0 (230 VAC) OR 4.0 (460 VAC)	Sets the motor overload current, here calculated to be the motor FLA x 1.15 S.F. and rounded up to the nearest .1 amps.
P034	Motor NP FLA	<u>1/3 HP</u> 2.3 (230V) OR 1.15 (460V)	<u>1/4 HP</u> 1.9 (230V) OR 1.0 (460V)	3.6 (230 VAC) OR 1.8 (460 VAC) OR 1.6 (575 VAC)	6.8 (230 VAC) OR 3.4 (460 VAC)	Use 230 values for 190-240 VAC applications. Use 460 values for 380-500 VAC applications Set to motor nameplate rated full load amps rounded up to nearest .1 amps.
P035	Motor Poles	6				Defines the number of poles in the motor. Default = 4
P036	Motor NP RPM	1200				Set to motor nominal RPM for 4-pole and 6-pole motors.
P037	Motor NP Power	<u>1/3 HP</u> .25	<u>1/4HP</u> .20	0.75	1.5	Set to motor nameplate rated power. (kW)
P039	Torque Perf Mode	0				The method of motor control used in the drive. 0 = V/Hz 1 = "SVC" (Default) 2 = "Economize" 3 = "Vector"
P041	Accel Time 1	0.5		0.5	1.0	Sets the rate of accel 1 for all speed increases. Default = 10.0s.
P042	Decel Time 1	0.5		0.5	1.5	Sets the rate of decel 1 for all speed decreases. Default = 10.0s.
P043	Minimum Freq	20		24	29	Sets the lowest frequency the drive will output.
P044	Maximum Freq	50		60	53	Sets the highest frequency the drive will output.
P045	Stop Mode	0				Determines the stopping mode used by the drive when a stop is initiated. 0 = Ramp, CF (Default) 1 = Coast, CF 2 = DC Brake, CF

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PF-523/525 Parameter Number	Parameter Name	50E / 90E Values		260E / 260E-G2 / 260E-G3 Values (1 HP Motor)	260E-G3 Plus Values (2 HP Motor)	Functional Description
P046	Start Source 1	2 (Hardwired) OR 5 (Ethernet/IP)				Selects the source of the speed reference to the drive. 1 = Keypad (Default) 2 = DigIn TrmBlk 3 = Serial/DSI 4 = Network Opt. 5 = Ethernet/IP (PF525 Only)
P047	Speed Ref 1	7 (Hardwired) OR 15 (Ethernet/IP)				Selects the source of the speed reference to the drive. 1 = Drive Pot (Default) 6 = 4-20 mA 7 = Preset Frequency 15 = Ethernet/IP (PF525 Only)
A410	Preset Freq 0	50		60	53	Provides an internal fixed speed command value. Default = 0 Hz. Note: uses Accel 1 & Decel 1
A411	Preset Freq 1	45		50	45	Provides an internal fixed speed command value. Default = 5 Hz. Note: uses Accel 1 & Decel 1
A412	Preset Freq 2	40				Provides an internal fixed speed command value. Default = 10Hz. Note: uses Accel 2 & Decel 2
A413	Preset Freq 3	50		60	53	Provides an internal fixed speed command value. Default = 20Hz. Note: uses Accel 2 & Decel 2
A434	DC Brake Time	0.5		0	0	Sets the amount of time DC brake current is "injected" into the motor. Default = 0.0s.
A435	DC Brake Level	0.1 (208/230 VAC) OR 0.2 (400/460 VAC)		0.1 (230 VAC) OR 0.2 (460 VAC) OR 0.2 (575 VAC)		Defines the maximum DC brake current level injected into the motor when "DC Brake" is selected as a stop mode. The DC braking voltage used in this function is created by a PWM algorithm and may not generate the smooth holding force needed for some applications.
A440	PWM Frequency	2 kHz				Sets the carrier frequency for the PWM output. Drive derating may occur at higher carrier frequencies. Default = 4kHz.
A442	Accel Time 2	0.5		0.5	1.0	Sets the rate of accel 2 for all speed increases. Default = 10.0s.
A443	Decel Time 2	0.5		0.5	1.5	Sets the rate of decel 2 for all speed decreases. Default = 10.0s.
A484	Current Lmt Val	<u>1/3 HP</u> 3.5 (230V) OR 1.8 (460V)	<u>1/4 HP</u> 2.9 (230V) OR 1.5 (460V)	5.4 (230 VAC) OR 2.7 (460 VAC) OR 2.4 (575 VAC)	10.2 (230 VAC) OR 5.1 (460 VAC)	Maximum output current allowed before current limiting occurs. Choose the value for the voltage in the application, here calculated to be the motor FLA x 1.5 and rounded up to the nearest .1 amps.

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PF- 523/525 Parameter Number	Parameter Name	50E / 90E Values	260E / 260E-G2 / 260E-G3 Values (1 HP Motor)	260E-G3 Plus Values (2 HP Motor)	Functional Description
A530	Boost Select	8 (Default)		7	Sets the boost voltage and redefines the V/Hz curve. 7 = 2.5% of P031 8 = 5.0% of P031 (Default)
A534	Maximum Voltage	150 (208/230 VAC) OR 300 (400/460 VAC)	130 (208/230 VAC) OR 260 (400/460 VAC) OR 325 (575 VAC)	90 (230 VAC) OR 180 (460 VAC)	Sets the highest voltage the drive will output.
A543	Start At PowerUp	1			Enables/disables a feature to issue a Start or Run command and automatically resume running at commanded speed after drive input power is restored. Default = 0 “Disabled”
T062	DigIn TermBlk 02	48			Programmable Digital Input. 0 = Not Used 48 = 2-Wire Forward (Default) 49 = 3-Wire Start Note: Only applies if P047=07 (Preset Frequency)
T063	DigIn TermBlk 03	0			Programmable Digital Input. 0 = Not Used 50 = 2-Wire Reverse (Default) 51 = 3-Wire Directional Note: Only applies if P047=07 (Preset Frequency)
T065	DigIn TermBlk 05	7			Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T066	DigIn TermBlk 06	7			Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T076	Relay Out1 Sel	2			Determines the operation of the programmable output relay. 0 = Ready/Fault (Default) 1 = At Frequency 2 = Motor Running 3 = Reverse 4 = Motor Overload
T105	Safety open En	1 (normal) OR 0 (if required by customer)			Sets the action when both safety inputs (Safety 1 and Safety 2) are disabled (de-energized – no power is applied). 0 = Fault Enable (Default) 1 = Fault Disable
IF ETHERNET IS USED SEE APPENDIX A FOR PARAMETERS.					

Table PH015-3 – PowerFlex 525 VFD Parameters for FastBack® Conveyor

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

Revolution Equipment

The following is a typical control I/O schematic diagram and drive settings for a Revolution Equipment.

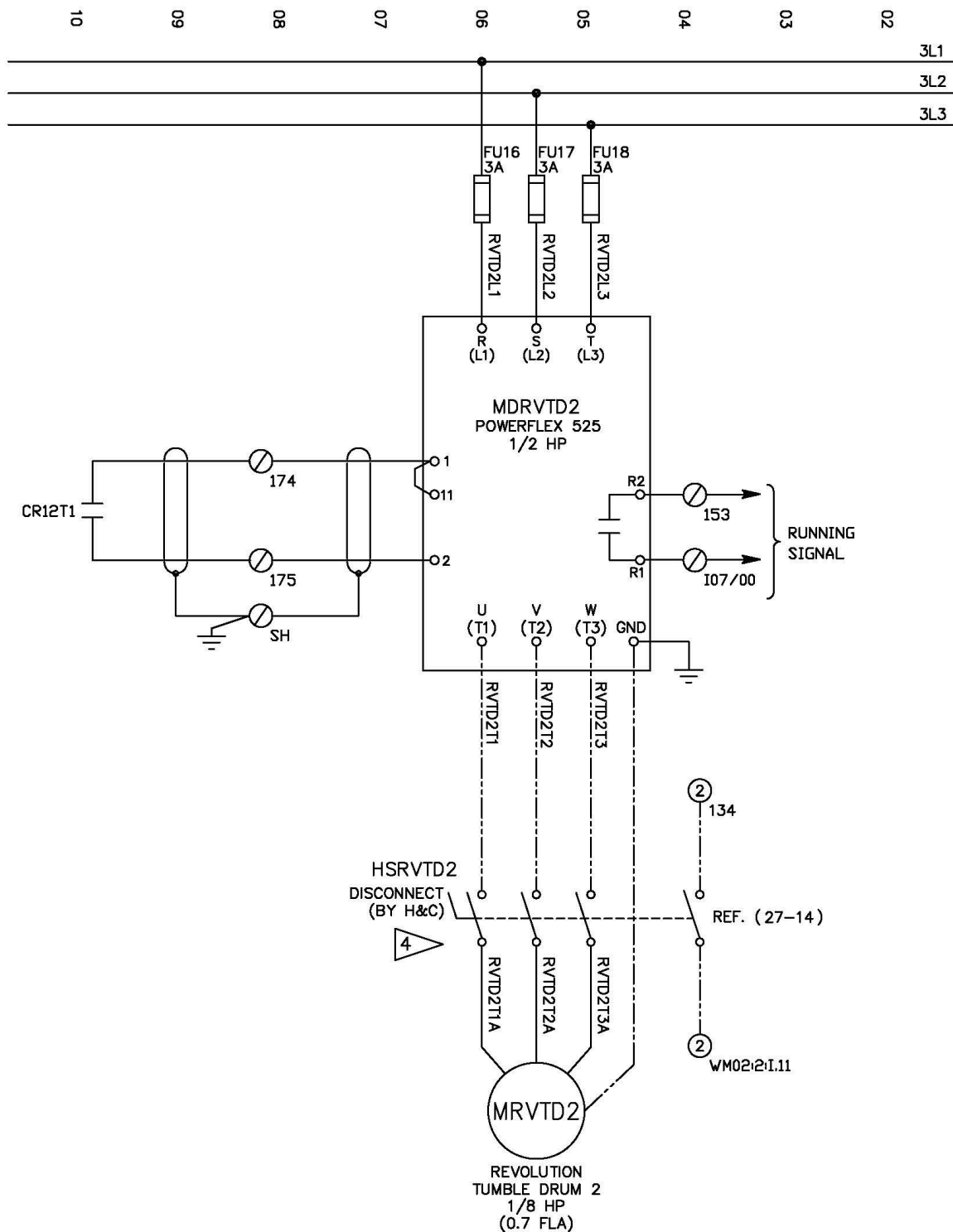


Figure PH015-5 – PowerFlex 525 Control I/O Wiring Block Diagram for a Tumble Drum

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 **SETUP PARAMETERS**

Heat and Control Product Handling Systems

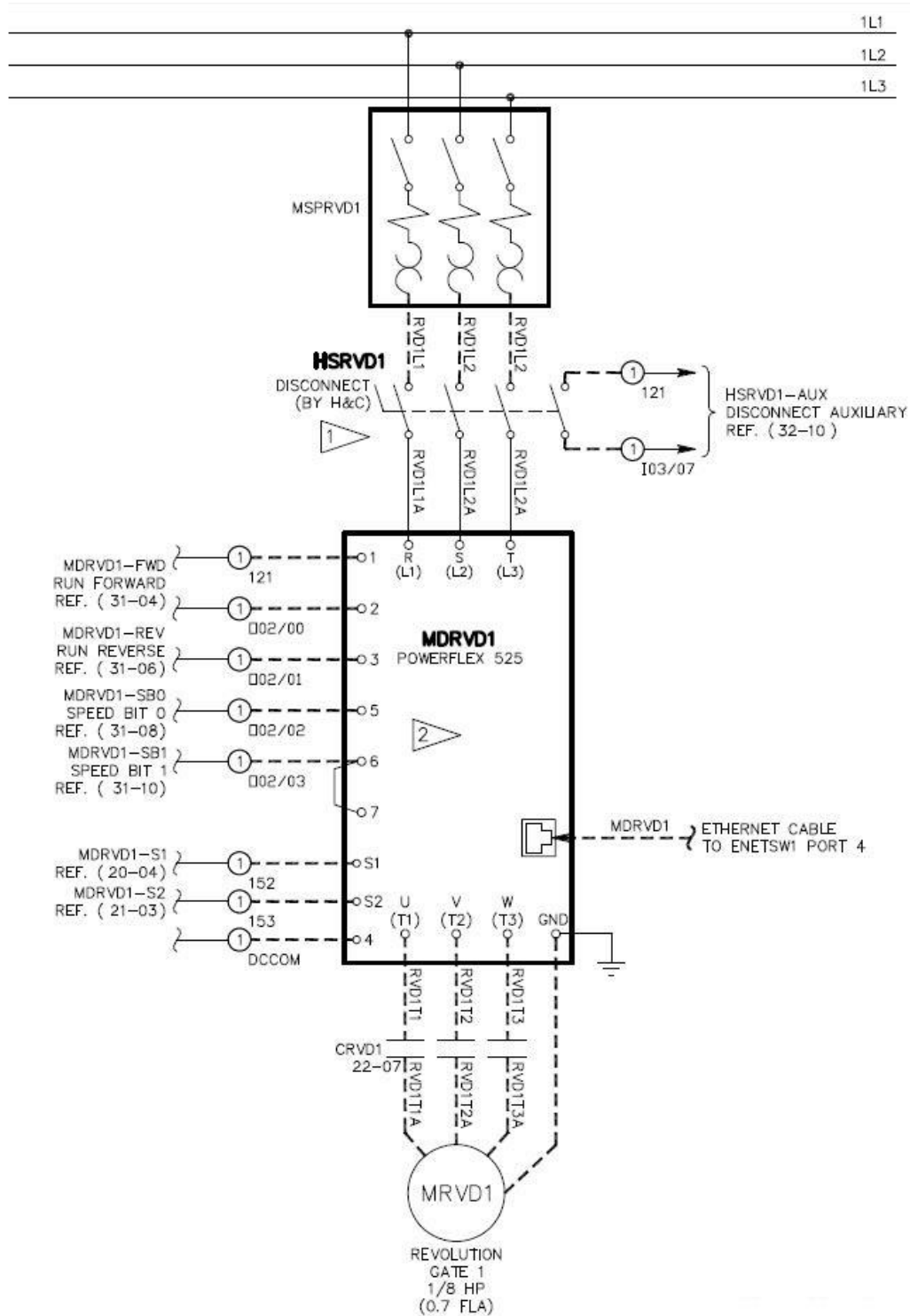


Figure PH015-6 – PowerFlex 525 Control I/O Wiring Block Diagram for a Rev. Gate

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

Allen-Bradley PowerFlex 525 Inverter Parameter Values for Revolution Equipment

PF-523/525 Parameter Number	Parameter Name	Generation 2.0 Revolution Tumble Drum (117RPM /15:1 Gearmotor)	Generation 2.0 Revolution Gate (70 RPM / 25:1 Gearmotor)	Function Description
P030	Language	1 "English"	1 "English"	Selects the display language when using an LCD HIM. This parameter is not functional with an LED HIM. Default = 1 "English"
P031	Motor NP Volts	208 OR 230 OR 400 Or 460		Set to line voltage
P032	Motor NP Hertz	60	60	Set to motor nameplate rated frequency.
P033	Motor OL Current	0.7 (230 VAC) OR 0.4 (460 VAC)	0.7 (230 VAC) OR 0.4 (460 VAC)	Sets the motor nameplate overload current.
P034	Motor NP FLA	0.7 (230 VAC) OR 0.4 (460 VAC)	0.7 (230 VAC) OR 0.4 (460 VAC)	Set to motor nameplate rated full load amps.
P035	Motor Poles	4	4	Defines the number of poles in the motor. Default = 4
P036	Motor NP RPM	1800	1800	Set to motor nominal RPM.
P037	Motor NP Power	0.12 (1/8HP)	0.12 (1/8HP)	Set to motor nameplate rated power. (kW)
P039	Torque Perf Mode	1	1	Selects the motor control mode. 0 = V/Hz 1 = SVC (Default) 2 = Economize 3 = Vector
P041	Accel Time 1	0.5	13.1 (Slow Speed)	Sets the rate of accel 1 for all speed increases. Default = 10.0s.
P042	Decel Time 1	0.5	0.2 (Slow Speed)	Sets the rate of decel 1 for all speed decreases. Default = 10.0s.
P043	Minimum Freq	13	13	Sets the lowest frequency the drive will output.
P044	Maximum Freq	80	67	Sets the highest frequency the drive will output.
P045	Stop Mode	0	0	Determines the stopping mode used by the drive when a stop is initiated. 0 = Ramp, CF (Default) 1 = Coast, CF 2 = DC Brake, CF
P046	Start Source 1	2 (Hardwired) OR 5 (Ethernet/IP)	2 (Hardwired) OR 5 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Keypad (Default) 2 = DigIn TrmBlk 3 = Serial/DSI 4 = Network Opt. 5 = Ethernet/IP
P047	Speed Ref 1	7 (Hardwired) OR 15 (Ethernet/IP)	7 (Hardwired) OR 15 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Drive Pot (Default) 6 = 4-20 mA 7 = Preset Frequency 15 = Ethernet/IP

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

PF-523/525 Parameter Number	Parameter Name	Generation 2.0 Revolution Tumble Drum (117RPM /15:1 Gearmotor)	Generation 2.0 Revolution Gate (70 RPM / 25:1 Gearmotor)	Function Description
A410	Preset Freq 0	80	n/a	Provides an internal fixed speed command value. Default = 0 Hz. Note: uses Accel 1 & Decel 1
A411	Preset Freq 1	n/a	13	Provides an internal fixed speed command value. Default = 5 Hz. Note: uses Accel 1 & Decel 1
A412	Preset Freq 2	n/a	67	Provides an internal fixed speed command value. Default = 10Hz. Note: uses Accel 2 & Decel 2
A440	PWM Frequency	2 kHz	2 kHz	Sets the carrier frequency for the PWM output. Drive derating may occur at higher carrier frequencies. Default = 4kHz.
A442	Accel Time 2	0.4	0.4 (Fast Speed)	Sets the rate of accel 2 for all speed increases. Default = 10.0s.
A443	Decel Time 2	0.1	0.1 (Fast Speed)	Sets the rate of decel 2 for all speed decreases. Default = 10.0s.
A484	Current Lmt Val	1.0 (230 VAC) OR 0.6 (460 VAC)	1.0 (230 VAC) OR 0.6 (460 VAC)	Maximum output current allowed before current limiting occurs. Choose the value for the voltage in the application
A543	Start At PowerUp	1	1	Enables/disables a feature to issue a Start or Run command and automatically resume running at commanded speed after drive input power is restored. Default = 0 "Disabled"
T062	DigIn TermBlk 02	48	48	Programmable Digital Input. 0 = Not Used 48 = 2-Wire Forward (Default) 49 = 3-Wire Start Note: Only applies if P047=07 (Preset Frequency)
T063	DigIn TermBlk 03	0	50	Programmable Digital Input. 0 = Not Used 50 = 2-Wire Reverse (Default) 51 = 3-Wire Directional Note: Only applies if P047=07 (Preset Frequency)
T065	DigIn TermBlk 05	7	7	Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T066	DigIn TermBlk 06	7	7	Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T067	DigIn TermBlk 07	N/A	11	Selects the function for the digital inputs. 11 = "Acc/Dec Sel2" 05 = "Spd + Strt 2" Default

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

PF-523/525 Parameter Number	Parameter Name	Generation 2.0 Revolution Tumble Drum (117RPM /15:1 Gearmotor)	Generation 2.0 Revolution Gate (70 RPM / 25:1 Gearmotor)	Function Description
T076	Relay Out1 Sel	2	2	Determines the operation of the programmable output relay. 0 = Ready/Fault (Default) 1 = At Frequency 2 = Motor Running 3 = Reverse 4 = Motor Overload
T105	Safety open En	1 (normal) OR 0 (if required by customer)	1 (normal) OR 0 (if required by customer)	Sets the action when both safety inputs (Safety 1 and Safety 2) are disabled (de-energized – no power is applied). 0 = Fault Enable (Default) 1 = Fault Disable
IF <u>ETHERNET</u> IS USED SEE APPENDIX A FOR PARAMETERS.				

Table PH015-4 – PowerFlex 525 Drive Parameters for Revolution Equipment

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

Switchback Accumulator

The following is a typical control I/O schematic diagram and drive settings for a Switchback Accumulator.

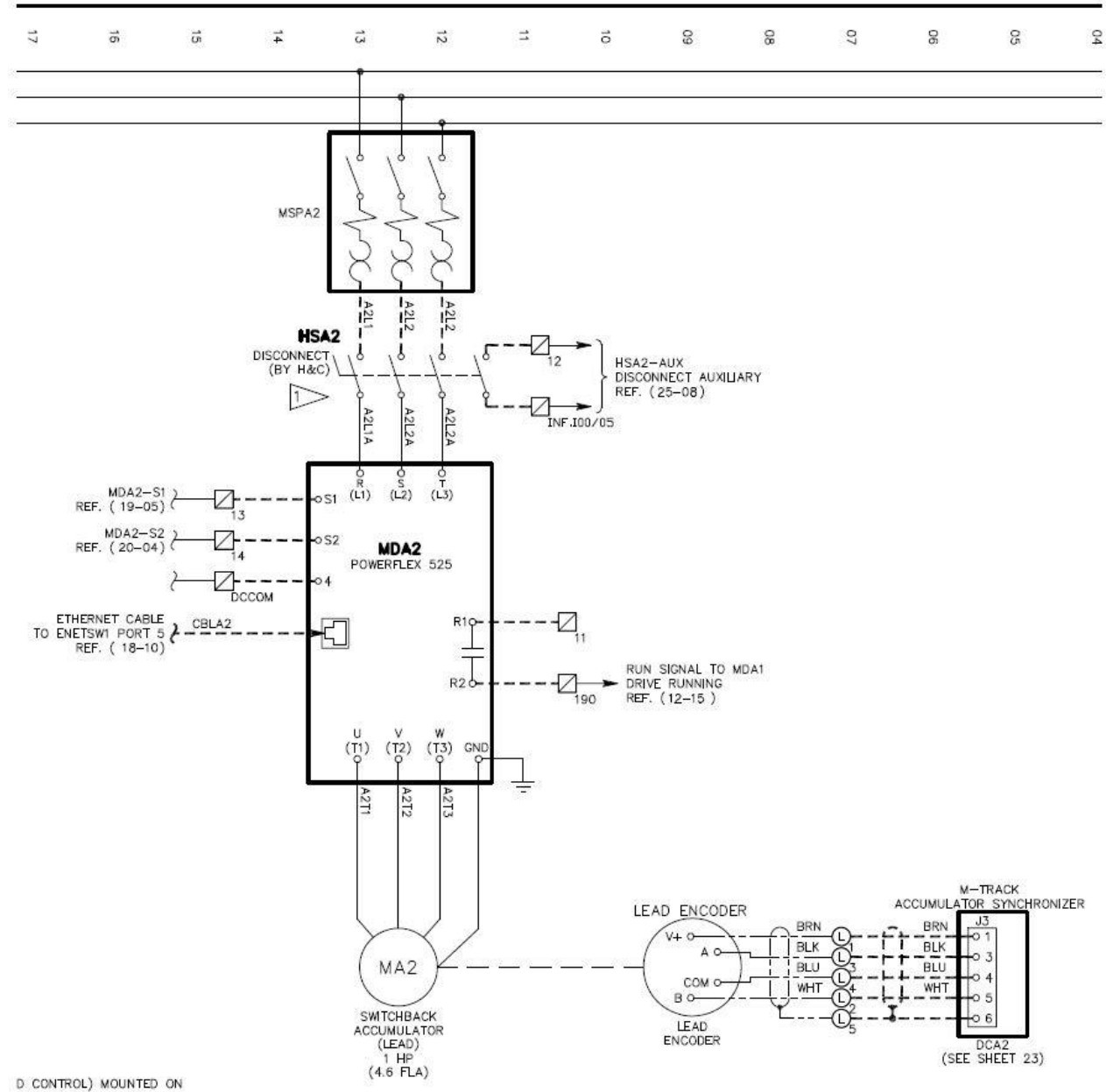


Figure PH015-7 – PowerFlex 525 Switchback Accumulator (Lead) Control I/O Wiring Block Diagram

Heat and Control Product Handling Systems



TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

Allen-Bradley PowerFlex 525 VFD Parameter Values for Switchback Accumulator

PF-523/525 Parameter Number	Parameter Name	Switchback Accumulator (Lead-260E) Values	Switchback Accumulator (Follower-260E) Values	Functional Description
P030	Language	1 "English"	1 "English"	Selects the display language when using an LCD HIM. This parameter is not functional with an LED HIM. Default = 1 "English"
P031	Motor NP Volts	208 OR 230 OR 400 OR 460		Set to line voltage
P032	Motor NP Hertz	60	60	Set to motor nameplate rated frequency.
P033	Motor OL Current	4.2 (230 VAC) OR 2.1 (460 VAC) OR 1.9 (575 VAC)	4.2 (230 VAC) OR 2.1 (460 VAC) OR 1.9 (575 VAC)	Sets the motor overload current, here calculated to be the motor FLA x 1.15 S.F. and rounded up to the nearest .1 amps.
P034	Motor NP FLA	3.6 (230 VAC) OR 1.8 (460 VAC) OR 1.6 (575 VAC)	3.6 (230 VAC) OR 1.8 (460 VAC) OR 1.6 (575 VAC)	Use 230 values for 190-240 VAC applications. Use 460 values for 380-500 VAC applications. Set to motor nameplate rated full load amps rounded up to nearest .1 amps.
P035	Motor Poles	6	6	Defines the number of poles in the motor. Default = 4
P036	Motor NP RPM	1200	1200	Set to motor nominal RPM.
P037	Motor NP Power	0.75 (1HP)	0.75 (1HP)	Set to motor nameplate rated power. (kW)
P039	Torque Perf Mode	0	0	The method of motor control used in the drive. 0 = V/Hz 1 = "SVC" .(Default) 2 = "Economize" 3 = "Vector"
P041	Accel Time 1	1.0	0.5	Sets the rate of accel 1 for all speed increases. Default = 10.0s.
P042	Decel Time 1	1.5	0.8	Sets the rate of decel 1 for all speed decreases. Default = 10.0s.
P043	Minimum Freq	28	0	Sets the lowest frequency the drive will output.
P044	Maximum Freq	50	60	Sets the highest frequency the drive will output.
P045	Stop Mode	0	0	Determines the stopping mode used by the drive when a stop is initiated 0 = Ramp, CF (Default) 1 = Coast, CF 2 = DC Brake, CF
P046	Start Source 1	2 (Hardwired) OR 5 (Ethernet/IP)	2 (Hardwired)	Selects the source of the speed reference to the drive. 1 = Keypad (Default) 2 = DigIn TrmBlk 3 = Serial/DSI 4 = Network Opt. 5 = Ethernet/IP
P047	Speed Ref 1	7 (Hardwired) OR 15 (Ethernet/IP)	5 (Hardwired)	Selects the source of the speed reference to the drive. 1 = Drive Pot (Default) 5 = 0-10 V Input 6 = 4-20 mA 7 = Preset Frequency 15 = Ethernet/IP
A410	Preset Freq 0	50	60	Provides an internal fixed speed command value. Default = 0 Hz. Note: uses Accel 1 & Decel 1

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525

SETUP PARAMETERS

Heat and Control Product Handling Systems

PF-523/525 Parameter Number	Parameter Name	Switchback Accumulator (Lead-260E) Values	Switchback Accumulator (Follower-260E) Values	Functional Description
A411	Preset Freq 1	50	60	Provides an internal fixed speed command value. Default = 5 Hz. Note: uses Accel 1 & Decel 1
A412	Preset Freq 2	40	50	Provides an internal fixed speed command value. Default = 10Hz. Note: uses Accel 2 & Decel 2
A413	Preset Freq 3	35	40	Provides an internal fixed speed command value. Default = 20Hz. Note: uses Accel 2 & Decel 2
A434	DC Brake Time	0.5	0.0	Sets the length of time that DC brake current is “injected” into the motor. Default = 0.0 Sec.
A435	DC Brake Level	0.16	NA	Defines the maximum DC brake current, in amps, applied to the motor when P045 [Stop Mode] is set to either 4 “Ramp” or 6 “DC Brake”. Default = Drive Rated Amps x 0.05
A440	PWM Frequency	2 kHz	2 kHz	Sets the carrier frequency for the PWM output. Drive derating may occur at higher carrier frequencies. Default = 4kHz.
A442	Accel Time 2	1.0	n/a	Sets the rate of accel 2 for all speed increases. Default = 10.0s.
A443	Decel Time 2	1.5	n/a	Sets the rate of decel 2 for all speed decreases. Default = 10.0s.
A484	Current Lmt Val	5.4 (230 VAC) OR 2.7 (460 VAC) OR 2.4 (575 VAC)	5.4 (230 VAC) OR 2.7 (460 VAC) OR 2.4 (575 VAC)	Maximum output current allowed before current limiting occurs. Choose the value for the voltage in the application, here calculated to be the motor FLA x 1.5 and rounded up to the nearest .1 amps.
A534	Maximum Voltage	130 (208/230 VAC) OR 260 (400/460 VAC) OR 325 (575 VAC)	130 (208/230 VAC) OR 260 (400/460 VAC) OR 325 (575 VAC)	Sets the highest voltage the drive will output.
A543	Start At PowerUp	1	1	Enables/disables a feature to issue a Start or Run command and automatically resume running at commanded speed after drive input power is restored. Default = 0 “Disabled”
A544	Reversed Disable	1	1	Enables/disables the function that allows the direction of motor rotation to be changed. 0 = Rev. Enabled (Default) 1 = Rev. Disabled
T062	DigIn TermBlk 02	48	48	Programmable Digital Input. 0 = Not Used 48 = 2-Wire Forward (Default) 49 = 3-Wire Start
T065	DigIn TermBlk 05	7	n/a	Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T066	DigIn TermBlk 06	7	n/a	Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T076	Relay Out1 Sel	2	2	Determines the operation of the programmable output relay. 0 = Ready/Fault (Default) 1 = At Frequency 2 = Motor Running 3 = Reverse 4 = Motor Overload

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

PF-523/525 Parameter Number	Parameter Name	Switchback Accumulator (Lead-260E) Values	Switchback Accumulator (Follower-260E) Values	Functional Description
T105	Safety open En	1 (normal) OR 0 (if required by customer)	1 (normal) OR 0 (if required by customer)	Sets the action when both safety inputs (Safety 1 and Safety 2) are disabled (de-energized – no power is applied). 0 = Fault Enable (Default) 1 = Fault Disable
IF ETHERNET IS USED SEE APPENDIX A FOR PARAMETERS.				

Table PH015-5 – PowerFlex 525 VFD Parameters for FastBack® Switchback Accumulator

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

Allen-Bradley PowerFlex 525 VFD Parameter Values for Tumble Drum Units

PF-523/525 Parameter Number	Parameter Name	Values for Stand-Alone TUMBLE DRUM UNITS	Function Description
P030	Language	1 "English"	Selects the display language when using an LCD HIM. This parameter is not functional with an LED HIM. Default = 1 "English"
P031	Motor NP Volts	208 OR 230 OR 400 OR 460	Set to line voltage
P032	Motor NP Hertz	60	Set to motor nameplate rated frequency.
P033	Motor OL Current	2.0 (230 VAC) OR 1.0 (460 VAC)	Sets the motor nameplate overload current
P034	Motor NP FLA	2.0 (230 VAC) OR 1.0 (460 VAC)	Set to motor nameplate rated full load amps.
P035	Motor Poles	4	Defines the number of poles in the motor. Default = 4
P036	Motor NP RPM	1800	Set to motor nominal RPM.
P037	Motor NP Power	0.4 (1/2HP)	Set to motor nameplate rated power. (kW)
P039	Torque Perf Mode	1	The method of motor control used in the drive. 0 = V/Hz 1 = "SVC" (Default) 2 = "Economize" 3 = "Vector"
P041	Accel Time 1	0.5	Sets the rate of accel 1 for all speed increases. Default = 10.0s.
P042	Decel Time 1	0.5	Sets the rate of decel 1 for all speed decreases. Default = 10.0s.
P043	Minimum Freq	0	Sets the lowest frequency the drive will output. Default = 0.00 Hz
P044	Maximum Freq	90	Sets the highest frequency the drive will output.
P045	Stop Mode	0	Determines the stopping mode used by the drive when a stop is initiated 0 = Ramp, CF (Default) 1 = Coast, CF 2 = DC Brake, CF
P046	Start Source 1	2 (Hardwired) OR 5 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Keypad (Default) 2 = DigIn TrmBlk 3 = Serial/DSI 4 = Network Opt. 5 = Ethernet/IP
P047	Speed Ref 1	7 (Hardwired) OR 15 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Drive Pot (Default) 6 = 4-20 mA 7 = Preset Frequency 15 = Ethernet/IP

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

PF-523/525 Parameter Number	Parameter Name	Values for Stand-Alone TUMBLE DRUM UNITS	Function Description
A410	Preset Freq 0	66 (for 28" dia. Drum w/ SST Flights) OR 75 (for 32" dia. Drum w/ SST Flights) OR 80 (for 34" dia. Barrels which accept the 28" and 32" dia. Removable liners) OR 90 (for 40" dia. Drum w/ SST Flights)	Provides an internal fixed speed command value. Default = 0 Hz. Note: uses Accel 1 & Decel 1
A440	PWM Frequency	2 kHz	Sets the carrier frequency for the PWM output. Drive derating may occur at higher carrier frequencies. Default = 4kHz.
A484	Current Lmt Val	3.0 (230 VAC) OR 1.5 (460 VAC)	Maximum output current allowed before current limiting occurs. Choose the value for the voltage in the application
A543	Start At PowerUp	1	Enables/disables a feature to issue a Start or Run command and automatically resume running at commanded speed after drive input power is restored. Default = 0 "Disabled"
T062	DigIn TermBlk 02	48	Programmable Digital Input. 0 = Not Used 48 = 2-Wire Forward (Default) 49 = 3-Wire Start Note: Only applies if P047=07 (Preset Frequency)
T076	Relay Out1 Sel	2	Determines the operation of the programmable output relay. 0 = Ready/Fault (Default) 1 = At Frequency 2 = Motor Running 3 = Reverse 4 = Motor OverLoad
T105	Safety open En	1 (normal) OR 0 (if required by customer)	Sets the action when both safety inputs (Safety 1 and Safety 2) are disabled (de- energized – no power is applied). 0 = Fault Enable (Default) 1 = Fault Disable
IF ETHERNET IS USED SEE APPENDIX A FOR PARAMETERS.			

Table PH015-6 – PowerFlex 525 Drive Parameters for Tumble Drum Units

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

Allen-Bradley PowerFlex 525 VFD Parameter Values for Incline / Belted Conveyors

PF-523/525 Parameter Number	Parameter Name	Values for INCLINE / BELTED CONVEYORS	Function Description
P030	Language	1 "English"	Selects the display language when using an LCD HIM. This parameter is not functional with an LED HIM. Default = 1 "English"
P031	Motor NP Volts	208 OR 230 OR 400 OR 460	Set to line voltage
P032	Motor NP Hertz	60	Set to motor nameplate rated frequency.
P033	Motor OL Current	Same as FLA	Sets the motor nameplate overload current
P034	Motor NP FLA	Varies	Set to motor nameplate rated full load amps.
P035	Motor Poles	4 or 6	Defines the number of poles in the motor. Default = 4
P036	Motor NP RPM	1800 or 1200	Set to motor nominal RPM.
P037	Motor NP Power	Varies	Set to motor nameplate rated power. (kW)
P039	Torque Perf Mode	1	The method of motor control used in the drive. 0 = V/Hz 1 = "SVC" (Default) 2 = "Econimize" 3 = "Vector"
P041	Accel Time 1	1.0	Sets the rate of accel 1 for all speed increases. Default = 10.0s.
P042	Decel Time 1	1.0	Sets the rate of decel 1 for all speed decreases. Default = 10.0s.
P043	Minimum Freq	0	Sets the lowest frequency the drive will output. Default = 0.00 Hz
P044	Maximum Freq	60	Sets the highest frequency the drive will output.
P045	Stop Mode	0	Determines the stopping mode used by the drive when a stop is initiated. 0 = Ramp, CF (Default) 1 = Coast, CF 2 = DC Brake, CF
P046	Start Source 1	2 (Hardwired) OR 5 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Keypad (Default) 2 = DigIn TrmBlk 3 = Serial/DSI 4 = Network Opt. 5 = Ethernet/IP
P047	Speed Ref 1	7 (Hardwired) OR 15 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Drive Pot (Default) 6 = 4-20 mA 7 = Preset Frequency 15 = Ethernet/IP
A410	Preset Freq 0	60 NOTE: Vary speed as required for best operation.	Provides an internal fixed speed command value. Default = 0 Hz. Note: uses Accel 1 & Decel 1

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

PF-523/525 Parameter Number	Parameter Name	Values for INCLINE / BELTED CONVEYORS	Function Description
A440	PWM Frequency	2 kHz	Sets the carrier frequency for the PWM output. Drive derating may occur at higher carrier frequencies. Default = 4kHz.
A484	Current Lmt Val	1.5 * FLA (230 VAC) OR 1.5 * FLA (460 VAC)	Maximum output current allowed before current limiting occurs. Choose the value for the voltage in the application
A543	Start At PowerUp	1	Enables/disables a feature to issue a Start or Run command and automatically resume running at commanded speed after drive input power is restored. Default = 0 "Disabled"
T062	DigIn TermBlk 02	48	Programmable Digital Input. 0 = Not Used 48 = 2-Wire Forward (Default) 49 = 3-Wire Start Note: Only applies if P047=07 (Preset Frequency)
T076	Relay Out1 Sel	2	Determines the operation of the programmable output relay. 0 = Ready/Fault (Default) 1 = At Frequency 2 = Motor Running 3 = Reverse 4 = Motor Overload
T105	Safety open En	1 (normal) OR 0 (if required by customer)	Sets the action when both safety inputs (Safety 1 and Safety 2) are disabled (de-energized – no power is applied). 0 = Fault Enable (Default) 1 = Fault Disable
IF ETHERNET IS USED SEE APPENDIX A FOR PARAMETERS.			

Table PH015-7 – PowerFlex 525 Drive Parameters for Incline / Belted Conveyors

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

Allen-Bradley PowerFlex 525 VFD Parameter Values for AC TecWeigh Seasoning Applicator (S5A)

PF-523/525 Parameter Number	Parameter Name	S5A Values (beyond the factory default settings)		Function Description
P030	Language	1 "English"		Selects the display language when using an LCD HIM. This parameter is not functional with an LED HIM. Default = 1 "English"
P031	Motor NP Volts	208 OR 230 OR 400 OR 460		Set to line voltage
P032	Motor NP Hertz	60		Set to motor nameplate rated frequency.
P033	Motor OL Current	Black Painted Motor	SST Motor	Sets the motor nameplate overload current.
		2.0 (208 VAC)	1.8 (208 VAC)	
		OR	OR	
		1.8 (230 VAC)	1.6 (230 VAC)	
		OR	OR	
P034	Motor NP FLA	Black Painted Motor	SST Motor	Set to motor nameplate rated full load amps.
		2.0 (208 VAC)	1.8 (208 VAC)	
		OR	OR	
		1.8 (230 VAC)	1.6 (230 VAC)	
		OR	OR	
P035	Motor Poles	Black Painted Motor	SST Motor	Defines the number of poles in the motor. Default = 4
		2.0 (208 VAC)	1.8 (208 VAC)	
		OR	OR	
		1.8 (230 VAC)	1.6 (230 VAC)	
		OR	OR	
P036	Motor NP RPM	Black Painted Motor	SST Motor	Set to motor nominal RPM.
		2.0 (208 VAC)	1.8 (208 VAC)	
		OR	OR	
		1.8 (230 VAC)	1.6 (230 VAC)	
		OR	OR	
P037	Motor NP Power	Black Painted Motor	SST Motor	Set to motor nameplate rated power. (kW)
		2.0 (208 VAC)	1.8 (208 VAC)	
		OR	OR	
		1.8 (230 VAC)	1.6 (230 VAC)	
		OR	OR	
P039	Torque Perf Mode	1		The method of motor control used in the drive. 0 = V/Hz 1 = "SVC" (Default) 2 = "Economize" 3 = "Vector"
P040	Autotune	1 ("0" will be displayed after Autotune routine is complete.)		0 = Ready/Idle (default) 1 = Static Autotune (Runs at next "Start" cmd) 2 = Rotate Autotune (<u>not</u> recommended) <u>Note:</u> "Start" must be pressed or a start cmd given to begin the routine. Value resets to "0" when routine is complete.
P041	Accel Time 1	1.2		Sets the rate of accel 1 for all speed increases. Default = 10.0s.
P042	Decel Time 1	1.2		Sets the rate of decel 1 for all speed decreases. Default = 10.0s.
P043	Minimum Freq	0		Sets the lowest frequency the drive will output. Default = 0.00 Hz
P044	Maximum Freq	110		Sets the highest frequency the drive will output.
P045	Stop Mode	0		Determines the stopping mode used by the drive when a stop is initiated. 0 = Ramp, CF (Default) 1 = Coast, CF 2 = DC Brake, CF

TECHNICAL PUBLICATION PH015 – ALLEN-BRADLEY PF525 SETUP PARAMETERS

Heat and Control Product Handling Systems

PF-523/525 Parameter Number	Parameter Name	S5A Values (beyond the factory default settings)		Function Description
P046	Start Source 1	2 (Hardwired) OR 5 (Ethernet/IP)		Selects the source of the speed reference to the drive. 1 = Keypad (Default) 2 = DigIn TrmBlk 3 = Serial/DSI 4 = Network Opt. 5 = Ethernet/IP
P047	Speed Ref 1	6 (4 – 20 mA) OR 15 (Ethernet/IP)		Selects the source of the speed reference to the drive. 1 = Drive Pot (Default) 6 = 4-20 mA 7 = Preset Frequency 15 = Ethernet/IP
A440	PWM Frequency	2 kHz		Sets the carrier frequency for the PWM output. Drive derating may occur at higher carrier frequencies. Default = 4kHz.
A484	Current Lmt Val	Black Painted Motor	SST Motor	Maximum output current allowed before current limiting occurs. Choose the value for the voltage in the application
		3.0 (208 VAC) OR 2.7 (230 VAC) OR 1.5 (400 VAC) OR 1.3 (460 VAC)	2.7 (208 VAC) OR 2.4 (230 VAC) OR 1.5 (400 VAC) OR 1.2 (460 VAC)	
A486	Shear Pin 1 Level	Black Painted Motor	SST Motor	Sets the value of current at which the shear pin fault occurs. (x.x amps)
		2.0 (208 VAC) OR 1.8 (230 VAC) OR 1.0 (400 VAC) OR 0.9 (460 VAC)	1.8 (208 VAC) OR 1.6 (230 VAC) OR 1.0 (400 VAC) OR 0.8 (460 VAC)	
A487	Shear Pin 1 Time	0.50		Sets the continuous time the drive must be at or above parameter A486 before fault occurs. (x.xx Seconds)
A543	Start At PowerUp	1		Enables/disables a feature to issue a Start or Run command and automatically resume running at commanded speed after drive input power is restored. Default = 0 “Disabled”
T062	DigIn TermBlk 02	48		Programmable Digital Input. 0 = Not Used 48 = 2-Wire Forward (Default) 49 = 3-Wire Start Note: Only applies if P047=07 (Preset Frequency)
T076	Relay Out1 Sel	2		Determines the operation of the programmable output relay. 0 = Ready/Fault (Default) 1 = At Frequency 2 = Motor Running 3 = Reverse 4 = Motor Overload
T105	Safety open En	1 (normal) OR 0 (if required by customer)		Sets the action when both safety inputs (Safety 1 and Safety 2) are disabled (de-energized – no power is applied). 0 = Fault Enable (Default) 1 = Fault Disable
IF ETHERNET IS USED SEE APPENDIX A FOR PARAMETERS.				

Table PH00 15-8 – PowerFlex 525 Drive Parameters for Tecweigh Seasoning Applicator

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Heat and Control Product Handling Systems

Allen-Bradley PowerFlex 525 VFD Parameter Values for Modulating Conveyors

PF-523/525 Parameter Number	Parameter Name	50E / 90E Values	Functional Description
P030	Language	1 "English"	Selects the display language when using an LCD HIM. This parameter is not functional with an LED HIM. Default = 1 "English"
P031	Motor NP Volts	208 OR 230 OR 400 OR 460	Set to line voltage
P032	Motor NP Hertz	60	Set to motor nameplate rated frequency.
P033	Motor OL Current	2.2 (230 VAC) OR 1.2 (460 VAC)	Sets the motor nameplate overload current, here calculated to be the motor FLA x 1.15 S.F. and rounded up to the nearest .1 amps.
P034	Motor NP FLA	1.9 (230 VAC) OR 1.0 (460 VAC)	Use 230 values for 190-240 VAC applications. Use 460 values for 380-500 VAC applications Set to motor nameplate rated full load amps rounded up to nearest .1 amps.
P035	Motor Poles	6	Defines the number of poles in the motor. Default = 4
P036	Motor NP RPM	1200	Set to motor nominal RPM.
P037	Motor NP Power	.25 (1/4HP)	Set to motor nameplate rated power. (kW)
P039	Torque Perf Mode	0	The method of motor control used in the drive. 0 = V/Hz 1 = "SVC" (Default) 2 = "Economize" 3 = "Vector"
P041	Accel Time 1	0.5	Sets the rate of accel 1 for all speed increases. Default = 10.0s.
P042	Decel Time 1	0.5	Sets the rate of decel 1 for all speed decreases. Default = 10.0s.
P043	Minimum Freq	20	Sets the lowest frequency the drive will output.
P044	Maximum Freq	50	Sets the highest frequency the drive will output.
P045	Stop Mode	0	Determines the stopping mode used by the drive when a stop is initiated. 0 = Ramp, CF (Default) 1 = Coast, CF 2 = DC Brake, CF
P046	Start Source 1	2 (Hardwired) OR 5 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Keypad (Default) 2 = DigIn TrmBlk 3 = Serial/DSI 4 = Network Opt. 5 = Ethernet/IP
P047	Speed Ref 1	6 (4-20 mA) OR 15 (Ethernet/IP)	Selects the source of the speed reference to the drive. 1 = Drive Pot (Default) 6 = 4-20 mA 7 = Preset Frequency 15 = Ethernet/IP
A434	DC Brake Time	0.5	Sets the amount of time DC brake current is "injected" into the motor. Default = 0.0s.
A435	DC Brake Level	0.1 (208/230 VAC) OR 0.2 (400/460 VAC)	Defines the maximum DC brake current level injected into the motor when "DC Brake" is selected as a stop mode. The DC braking voltage used in this function is created by a PWM algorithm and may not generate the smooth holding force needed for some applications.
A440	PWM Frequency	2 kHz	Sets the carrier frequency for the PWM output. Drive derating may occur at higher carrier frequencies. Default = 4kHz.

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PF-523/525 Parameter Number	Parameter Name	50E / 90E Values	Functional Description
A484	Current Lmt Val	2.9 (230 VAC) OR 1.5 (460 VAC)	Maximum output current allowed before current limiting occurs. Choose the value for the voltage in the application, here calculated to be the motor FLA x 1.5 and rounded up to the nearest .1 amps.
A534	Maximum Voltage	150 (208/230 VAC) OR 300 (400/460 VAC)	Sets the highest voltage the drive will output.
A543	Start At PowerUp	1	Enables/disables a feature to issue a Start or Run command and automatically resume running at commanded speed after drive input power is restored. Default = 0 "Disabled"
T062	DigIn TermBlk 02	48	Programmable Digital Input. 0 = Not Used 48 = 2-Wire Forward (Default) 49 = 3-Wire Start Note: Only applies if P047=07 (Preset Frequency)
T063	DigIn TermBlk 03	0	Programmable Digital Input. 0 = Not Used 50 = 2-Wire Reverse (Default) 51 = 3-Wire Directional Note: Only applies if P047=07 (Preset Frequency)
T065	DigIn TermBlk 05	0	Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T066	DigIn TermBlk 06	0	Programmable Digital Input. 0 = Not Used 7 = Preset Frequency (Default)
T076	Relay Out1 Sel	2	Determines the operation of the programmable output relay. 0 = Ready/Fault (Default) 1 = At Frequency 2 = Motor Running 3 = Reverse 4 = Motor Overload
T105	Safety open En	1 (normal) OR 0 (if required by customer)	Sets the action when both safety inputs (Safety 1 and Safety 2) are disabled (de-energized – no power is applied). 0 = Fault Enable (Default) 1 = Fault Disable
IF ETHERNET IS USED SEE APPENDIX A FOR PARAMETERS.			

Table PH00 15-9 – PowerFlex 525 VFD Parameters for FastBack® Conveyor

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Appendix A – Ethernet Communication

(PF525 drives only)

PF-523/525 Parameter Number	Parameter Name	Values	Functional Description
C128	EN Addr. Sel.	1	Enables the IP Address, Subnet mask, and Gateway address. 1 = Parameters 2 = "BOOTP" (Default)
C129	EN IP Addr. Cfg 1	192	Sets the bytes in the IP Address. (i.e. – If the IP addr. is 192.168.100.31 then set as shown)
C130	EN IP Addr. Cfg 2	168	
C131	EN IP Addr. Cfg 3	100	
C132	EN IP Addr. Cfg 4	31	
C133	EN Subnet. Cfg 1	255	Sets the bytes in the Subnet Mask. (i.e. – If the subnet mask is 255.255.255.0 then set as shown)
C134	EN Subnet. Cfg 2	255	
C135	EN Subnet. Cfg 3	255	
C136	EN Subnet. Cfg 4	0	
C141	EN Rate Cfg	3 (or as required by customer)	Sets the network data rate at which EtherNet/IP communicates. 0 = "Auto detect" (Default) 1 = "10Mbps Full" 2 = "10Mbps Half" 3 = "100Mbps Full" 4 = "100Mbps Half"
C143	EN Comm Flt Actn	1	Sets the action that the EtherNet/IP interface and drive takes if the EtherNet/IP interface detects that Ethernet communications have been disrupted. 0 = "Fault" (Default) 1 = "Stop" 2 = "Zero Data" 3 = "Hold Last" 4 = "Send Flt Cfg"
C144	EN Idle Flt Actn	1	Sets the action that the EtherNet/IP interface and drive takes if the EtherNet/IP interface detects that the scanner is idle because the controller was switched to program mode. 0 = "Fault" (Default) 1 = "Stop" 2 = "Zero Data" 3 = "Hold Last" 4 = "Send Flt Cfg"
C157**	EN Data Out 1	43	Datalink parameter number whose value is read from the embedded EtherNet/IP data table. 0 = None (Default) 43 = Min Frequency
C158**	EN Data Out 2	44	Datalink parameter number whose value is read from the embedded EtherNet/IP data table. 0 = None (Default) 44 = Maximum Frequency

**Parameters C157 and C158 are used only if required.

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Revision Record

Revision Level	Date	By	Description
Revision 00	06/20/2014	Rich Rees	• Set Up Parameters for the FastBack Model FDX
Revision 01	09/01/2014	Scott Le	• Set Up Parameters for 90E's, 260E's, Rev. Gates, Tumble Drums, Seasoning Augers, and others
Revision 02	11/07/2014	Rich Rees	• Finalized first official PF525 Tech Pub
Revision 03	02/06/2015	Tom Knodell	• Updated parameter values P033, P034, P036 & A484 dealing with current values and motor rpm on all 6-pole motors to correct a gross over speeding condition. • Changed P039 to V/f control mode for the 260E. • Added P040=1 Static Autotune to the S5A Parameters. • Revised the speed and accel parameters for the FDX unit
Revision 04	04/15/2015	Rich Rees	• Changed all Motor NP RPM settings (P036) to 1800 (4 pole) or 1200 (6 pole). These nominal value will keep the slip compensation from adversely adding to the requested frequency speed. Solves an AB bug in the drive firmware. • Added "Attention" note for line/load reactor need and load reactor selection table.
Revision 05	01/04/2016	Ken Petri & Arlene Mason	• Added table of contents.
Revision 06	3/14/16	Ken Petri & Arlene Mason	• Replaced obsolete 200A information with 260E-G3 / 260E-G3 Plus information. • Revised 90E parameters to include 1/3 HP motor option.
Revision 07	06/14/16	Ken Petri	• Revised P031 to reflect actual voltage and prevent over current situations at low Hz (when actual = 208v, but P031 = 230v) • Removed all reference to 523